Current model (ordinary-matter shell + CoE core) vs. A_F, term by term

worked sheet, 27 Sep 2026

1. Model and where each term comes from. Sedov–Taylor flow $\vec{V} = k\,\vec{x}$ (position from the CoE; Taylor 1950; Sedov 1959), decelerated by a point core M_c at the CoE and a uniform shell Ω_s between $R_{\rm in}$ and $R_{\rm out}$ (fixed in space; Earth and all sources inside it). Spherical symmetry makes the field radial, $\vec{V} = h(q)\,\vec{q}$, with $q = s/D$ (D = CoE–Earth distance). Linear perturbation of the code's integration (from $\tau_i = 0.01$, $Ht \simeq 1$):

$$\frac{h(q)}{k} \simeq 1 - \tfrac{1}{2}\Big[\underbrace{\tfrac{M_c}{M_{\rm crit}(D)}\big(\tfrac{1}{\tau_i}-1\big)q^{-3}}_{\text{core tidal term}} + \Omega_s\Big(\underbrace{\tfrac{1}{2}}_{\text{uniform: } m=0 \text{ only}} \underbrace{-\tfrac{3}{2}r_{\rm in}^2 q^{-2} + r_{\rm in}^3 q^{-3}}_{\text{hole } (r_{\rm in}=R_{\rm in}/D)}\Big)\Big], \qquad \beta \equiv \frac{h'}{h}\Big|_{q=1}, \quad \gamma \equiv \frac{h''}{h}\Big|_{q=1}.$$

Chris's A: $a_f = |\Delta\vec{V}|$/separation with sources at $R = 14.3$ Gpc, $\varepsilon \equiv R/D$, $\mu = \cos b_0 \cos\psi$, $\psi = \ell - \ell_0$ (ℓ_0 = anti-CoE direction). Exactly $a_f \propto |h(q_s)(\hat{x} + \varepsilon\hat{n}) - h(1)\hat{x}|/\varepsilon$; to first order in ε (the higher orders are the r/D terms):

$$a_f^2 \propto 1 + B\mu^2 + \varepsilon\,(P\mu + Q\mu^3), \qquad B = 2\beta + \beta^2, \quad P = \beta(3+\beta), \quad Q = (1+\beta)(\gamma+\beta) - 2\beta.$$

Take the square root, then use $\cos^2\psi = \frac{1}{2}(1+\cos 2\psi)$, $\cos^3\psi = \frac{1}{4}(3\cos\psi + \cos 3\psi)$, $\cos^4\psi = \frac{3}{8} + \frac{1}{2}\cos 2\psi + \frac{1}{8}\cos 4\psi$, $\cos^5\psi = \frac{1}{16}(10\cos\psi + 5\cos 3\psi + \cos 5\psi)$ ($b_0 = 0$ as in the code):

$$\frac{a_f}{\langle a_f\rangle} = 1 + a_1\cos\psi + a_2\cos 2\psi + a_3\cos 3\psi + \ldots, \quad a_2 = \frac{\frac{B}{4} - \frac{B^2}{16}}{m_0}, \quad a_1 = \frac{\varepsilon}{2m_0}\Big[P + \tfrac{3}{4}\big(Q - \tfrac{PB}{2}\big) - \tfrac{5QB}{16}\Big], \quad a_3 = \frac{\varepsilon}{2m_0}\Big[\tfrac{1}{4}\big(Q - \tfrac{PB}{2}\big) - \tfrac{5QB}{32}\Big],$$

with $m_0 = 1 + \frac{B}{4} - \frac{3B^2}{64}$. For a pure point mass ($\gamma = -4\beta$, $\beta \to 3K$) this is the earlier line-of-sight form $K[(3\mu^2-1) + \varepsilon(\frac{9}{2}\mu - \frac{15}{2}\mu^3)]$. The geometry gives only $\cos m\psi$ terms: even harmonics set by β, odd ones by εQ. *Rotation* (the sine factor, $A = a_f[1 - \mathrm{Amp}\sin(\psi+\varphi)]$) adds $C_1 = a_1 - \mathrm{Amp}\sin\varphi(1 + \frac{a_2}{2})$, $S_1 = -\mathrm{Amp}\cos\varphi(1 - \frac{a_2}{2})$, $C_3 = a_3 - \frac{\mathrm{Amp}\,a_2}{2}\sin\varphi$, $S_3 = -\frac{\mathrm{Amp}\,a_2}{2}\cos\varphi$ (C_m, S_m: $\cos m\psi, \sin m\psi$ parts). **Latitude:** the code evaluates the model in the Galactic plane only (one direction per degree, no $|b|$ cut, no sector average), so the match is done the same way. A tilt b_0 would give $B \to B\cos^2 b_0$ and ε-terms $\propto \cos b_0, \cos^3 b_0$, degenerate with β and εQ; therefore $b_0 = 0$.

2. Term-by-term solution (Newtonian, series above, point-core shape $\gamma = -4\beta$; exact least-squares in brackets):

Term	A_F coefficient	Analytic expression	Solved value
2ℓ phase	$\ell_2 = 14.96°$	ℓ_0 (axis; $\beta > 0$: fast along it)	$\ell_0 = 14.96°$ (12.8°)
2ℓ amplitude	$C_2 = 0.0734$	$a_2(\beta)$	$\beta = 0.161$ (0.210)
1ℓ (rotation)	$C_1 = 0.0398$, $S_1 = 0.0343$	C_1, S_1 of Sec. 1	Amp $= -0.068$, $\varphi = 1.02$ (-0.072, 1.03)
3ℓ cos	$C_3 = -0.0292$	$a_3(\varepsilon Q) - \frac{\mathrm{Amp}\,a_2}{2}\sin\varphi$	$\varepsilon Q = -0.308 \Rightarrow \varepsilon = 0.348$, $D = 41$ Gpc (47)
3ℓ sin	$S_3 = 0.0130$	$-\frac{\mathrm{Amp}\,a_2}{2}\cos\varphi$ (check)	predicted 0.0013: **not produced**
$m = 0$	$A_0 = 0.9965$	$N[1 - \frac{1}{2}\mathrm{Amp}\,a_1\sin\varphi]$	$N = 0.9971$
mass	–	$\beta \simeq \frac{3K}{1-K}$, $K = \frac{1}{2}\frac{M_c}{M_{\rm crit}(D)}(\frac{1}{\tau_i} - 1)$	$M_c/M_{\rm crit}(D) \approx 0.0010$ (linear; ≈0.0017 nonlinear)